

Customer_shipping_behaviour

October 6, 2026

```
[18]: import pandas as pd
df=pd.read_csv('customer_shopping_behavior.csv')
```

```
[7]: df.head()
```

```
[7]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	\
0	1	55	Male	Blouse	Clothing	53	
1	2	19	Male	Sweater	Clothing	64	
2	3	50	Male	Jeans	Clothing	73	
3	4	21	Male	Sandals	Footwear	90	
4	5	45	Male	Blouse	Clothing	49	

	Location	Size	Color	Season	Review Rating	Subscription Status	\
0	Kentucky	L	Gray	Winter	3.1	Yes	
1	Maine	L	Maroon	Winter	3.1	Yes	
2	Massachusetts	S	Maroon	Spring	3.1	Yes	
3	Rhode Island	M	Maroon	Spring	3.5	Yes	
4	Oregon	M	Turquoise	Spring	2.7	Yes	

	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	\
0	Express	Yes	Yes	14	
1	Express	Yes	Yes	2	
2	Free Shipping	Yes	Yes	23	
3	Next Day Air	Yes	Yes	49	
4	Free Shipping	Yes	Yes	31	

	Payment Method	Frequency of Purchases
0	Venmo	Fortnightly
1	Cash	Fortnightly
2	Credit Card	Weekly
3	PayPal	Weekly
4	PayPal	Annually

```
[6]: #Checking info
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
```

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	object
3	Item Purchased	3900 non-null	object
4	Category	3900 non-null	object
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	object
7	Size	3900 non-null	object
8	Color	3900 non-null	object
9	Season	3900 non-null	object
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	object
12	Shipping Type	3900 non-null	object
13	Discount Applied	3900 non-null	object
14	Promo Code Used	3900 non-null	object
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	object
17	Frequency of Purchases	3900 non-null	object

dtypes: float64(1), int64(4), object(13)

memory usage: 548.6+ KB

```
[8]: df.describe(include="all")
```

```
[8]:
```

	Customer ID	Age	Gender	Item Purchased	Category	\
count	3900.000000	3900.000000	3900	3900	3900	
unique	NaN	NaN	2	25	4	
top	NaN	NaN	Male	Blouse	Clothing	
freq	NaN	NaN	2652	171	1737	
mean	1950.500000	44.068462	NaN	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	NaN	
75%	2925.250000	57.000000	NaN	NaN	NaN	
max	3900.000000	70.000000	NaN	NaN	NaN	

	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	\
count	3900.000000	3900	3900	3900	3900	3863.000000	
unique	NaN	50	4	25	4	NaN	
top	NaN	Montana	M	Olive	Spring	NaN	
freq	NaN	96	1755	177	999	NaN	
mean	59.764359	NaN	NaN	NaN	NaN	3.750065	
std	23.685392	NaN	NaN	NaN	NaN	0.716983	
min	20.000000	NaN	NaN	NaN	NaN	2.500000	

25%	39.000000	NaN	NaN	NaN	NaN	3.100000
50%	60.000000	NaN	NaN	NaN	NaN	3.800000
75%	81.000000	NaN	NaN	NaN	NaN	4.400000
max	100.000000	NaN	NaN	NaN	NaN	5.000000

	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	\
count	3900	3900	3900	3900	3900
unique	2	6	2	2	2
top	No	Free Shipping	No	No	No
freq	2847	675	2223	2223	2223
mean	NaN	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN

	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900	3900
unique	NaN	6	7
top	NaN	PayPal	Every 3 Months
freq	NaN	677	584
mean	25.351538	NaN	NaN
std	14.447125	NaN	NaN
min	1.000000	NaN	NaN
25%	13.000000	NaN	NaN
50%	25.000000	NaN	NaN
75%	38.000000	NaN	NaN
max	50.000000	NaN	NaN

```
[10]: 0#Checking missing values
df.isnull().sum()
```

```
[10]: Customer ID      0
Age                  0
Gender              0
Item Purchased      0
Category            0
Purchase Amount (USD) 0
Location            0
Size                0
Color               0
Season              0
Review Rating       37
Subscription Status  0
Shipping Type       0
```

```
Discount Applied      0
Promo Code Used      0
Previous Purchases    0
Payment Method        0
Frequency of Purchases 0
dtype: int64
```

To impute the missing values by median of each item categories and replace by its own item by this we can avoid the bias over other

```
[19]: df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda x:
    ↪x.fillna(x.median()))
```

```
[20]: df.isnull().sum()
```

```
[20]: Customer ID      0
Age                  0
Gender              0
Item Purchased      0
Category            0
Purchase Amount (USD) 0
Location            0
Size                0
Color               0
Season              0
Review Rating       0
Subscription Status 0
Shipping Type       0
Discount Applied    0
Promo Code Used     0
Previous Purchases  0
Payment Method      0
Frequency of Purchases 0
dtype: int64
```

```
[21]: #Column names are not following any case, so making to snake case
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(" ", "_")
df=df.rename(columns={"purchase_amount_(usd)": "purchase_amount"})
```

```
[22]: df.columns
```

```
[22]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
    'purchase_amount', 'location', 'size', 'color', 'season',
    'review_rating', 'subscription_status', 'shipping_type',
    'discount_applied', 'promo_code_used', 'previous_purchases',
    'payment_method', 'frequency_of_purchases'],
    dtype='object')
```

Feature Engineering

```
[23]: #Creating column age group using qcut (Splits the data into q splits)
labels=['Young Adult','Adult','Middle-aged','senior']
df['age_group']=pd.qcut(df['age'],q=4,labels=labels)
```

```
[26]: df[['age','age_group']].head(10)
```

```
[26]:   age  age_group
0   55  Middle-aged
1   19  Young Adult
2   50  Middle-aged
3   21  Young Adult
4   45  Middle-aged
5   46  Middle-aged
6   63      senior
7   27  Young Adult
8   26  Young Adult
9   57  Middle-aged
```

```
[35]: #Create column purchase_frequency_Days
frequency_mapping={
    'Fortnightly':14,
    'Monthly':30,
    'Weekly':7,
    'Annually':365,
    'Bi-Weekly':14,
    'Quarterly':90,
    'Every 3 Months':90
}
df['purchase_frequency_days']=df['frequency_of_purchases'].
    ↪map(frequency_mapping)
```

```
[37]: df[['frequency_of_purchases','purchase_frequency_days']].head(10)
```

```
[37]:   frequency_of_purchases  purchase_frequency_days
0      Fortnightly           14
1      Fortnightly           14
2         Weekly            7
3         Weekly            7
4        Annually          365
5         Weekly            7
6      Quarterly           90
7         Weekly            7
8        Annually          365
9      Quarterly           90
```

```
[38]: #Does these two columns are same? There may be 2 scenarios  
#1.If Promo code used discount will be applied  
#2.If any promotional sale/ subscription is there discount are applied  
df[['discount_applied', 'promo_code_used']].head(10)
```

```
[38]: discount_applied promo_code_used  
0          Yes          Yes  
1          Yes          Yes  
2          Yes          Yes  
3          Yes          Yes  
4          Yes          Yes  
5          Yes          Yes  
6          Yes          Yes  
7          Yes          Yes  
8          Yes          Yes  
9          Yes          Yes
```

```
[40]: #checks whether both columns are same or not  
(df['discount_applied']== df['promo_code_used']).all()
```

```
[40]: True
```

The answer came as true so both are same, we can drop one column

```
[42]: df=df.drop('promo_code_used',axis=1)
```

```
[43]: df.columns
```

```
[43]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
        'purchase_amount', 'location', 'size', 'color', 'season',  
        'review_rating', 'subscription_status', 'shipping_type',  
        'discount_applied', 'previous_purchases', 'payment_method',  
        'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],  
       dtype='object')
```

```
[44]: pip install psycpg2-binary sqlalchemy
```

```
Requirement already satisfied: psycpg2-binary in  
c:\users\renuk\anaconda3\lib\site-packages (2.9.12)  
Requirement already satisfied: sqlalchemy in c:\users\renuk\anaconda3\lib\site-  
packages (2.0.34)  
Requirement already satisfied: typing-extensions>=4.6.0 in  
c:\users\renuk\anaconda3\lib\site-packages (from sqlalchemy) (4.11.0)  
Requirement already satisfied: greenlet!=0.4.17 in  
c:\users\renuk\anaconda3\lib\site-packages (from sqlalchemy) (3.0.1)  
Note: you may need to restart the kernel to use updated packages.
```

```
[45]: from sqlalchemy import create_engine

# Step 1: Connect to PostgreSQL
# Replace placeholders with your actual details
username = "postgres"      # default user
password = "rkrjs5" # the password you set during installation
host = "localhost"        # if running locally
port = "5433"              # default PostgreSQL port
database = "customer_behaviour" # the database you created in pgAdmin

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:
↳{port}/{database}")

# Step 2: Load DataFrame into PostgreSQL
table_name = "customer"    # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database_
↳'{database}'.")
```

Data successfully loaded into table 'customer' in database 'customer_behaviour'.